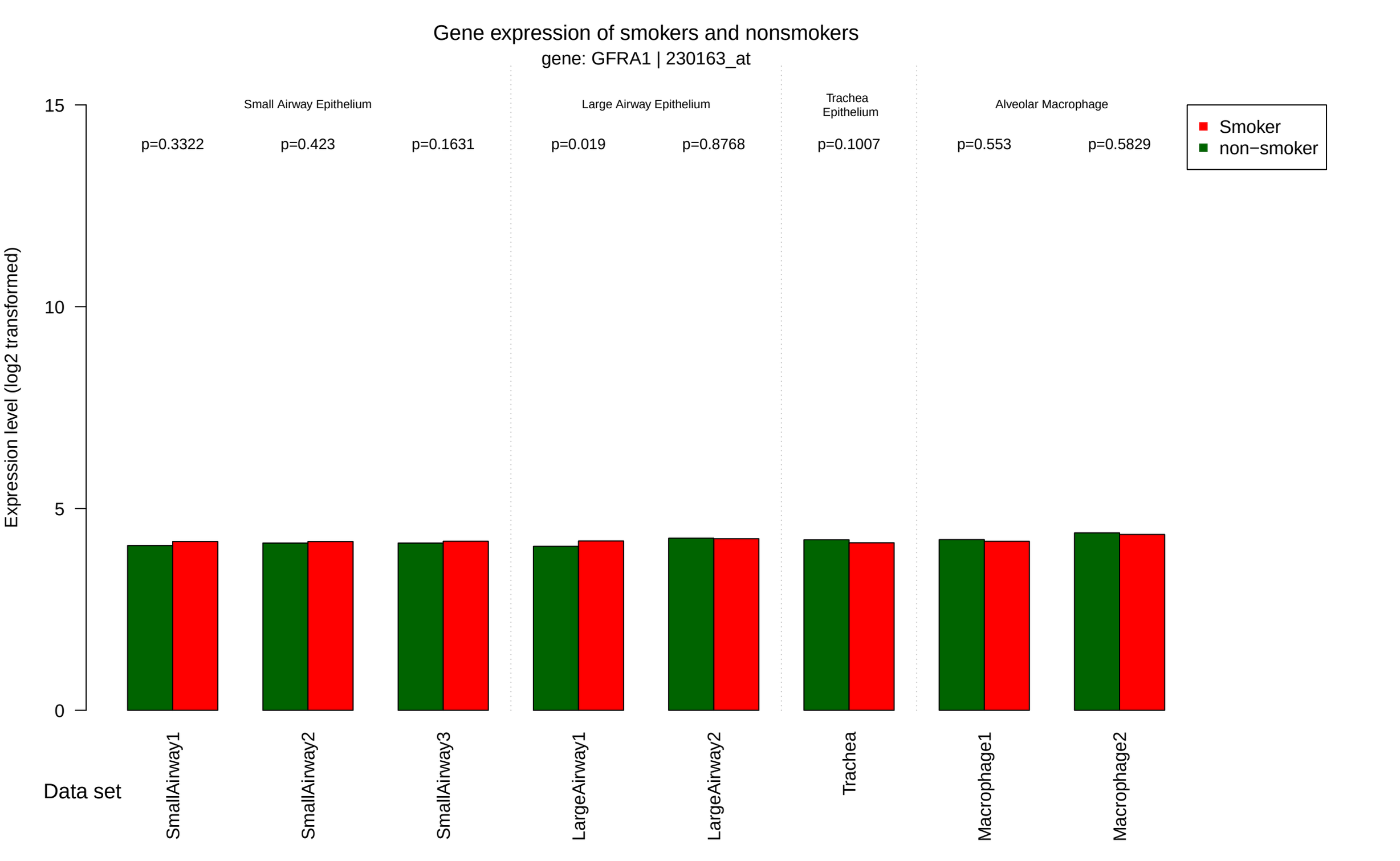
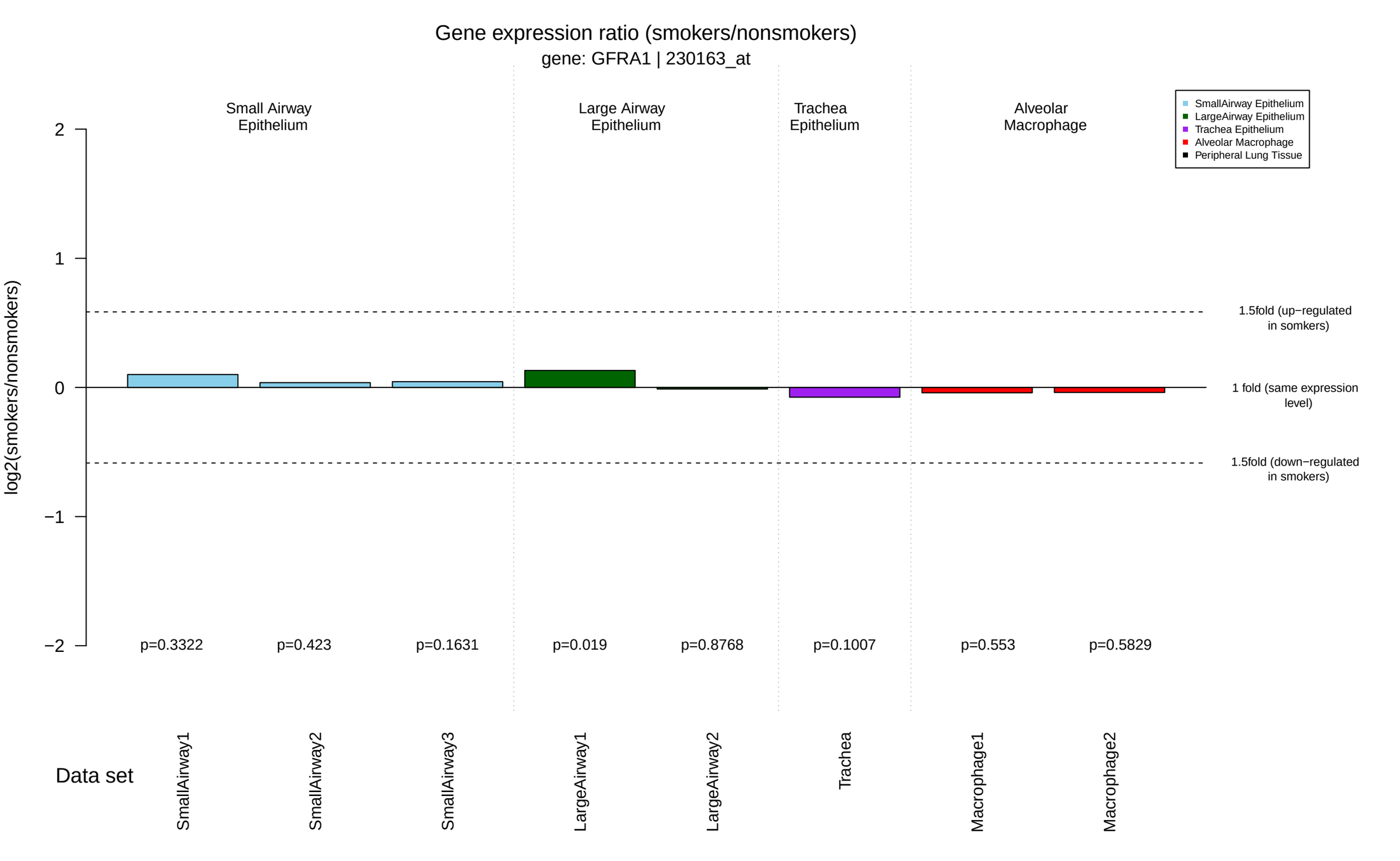
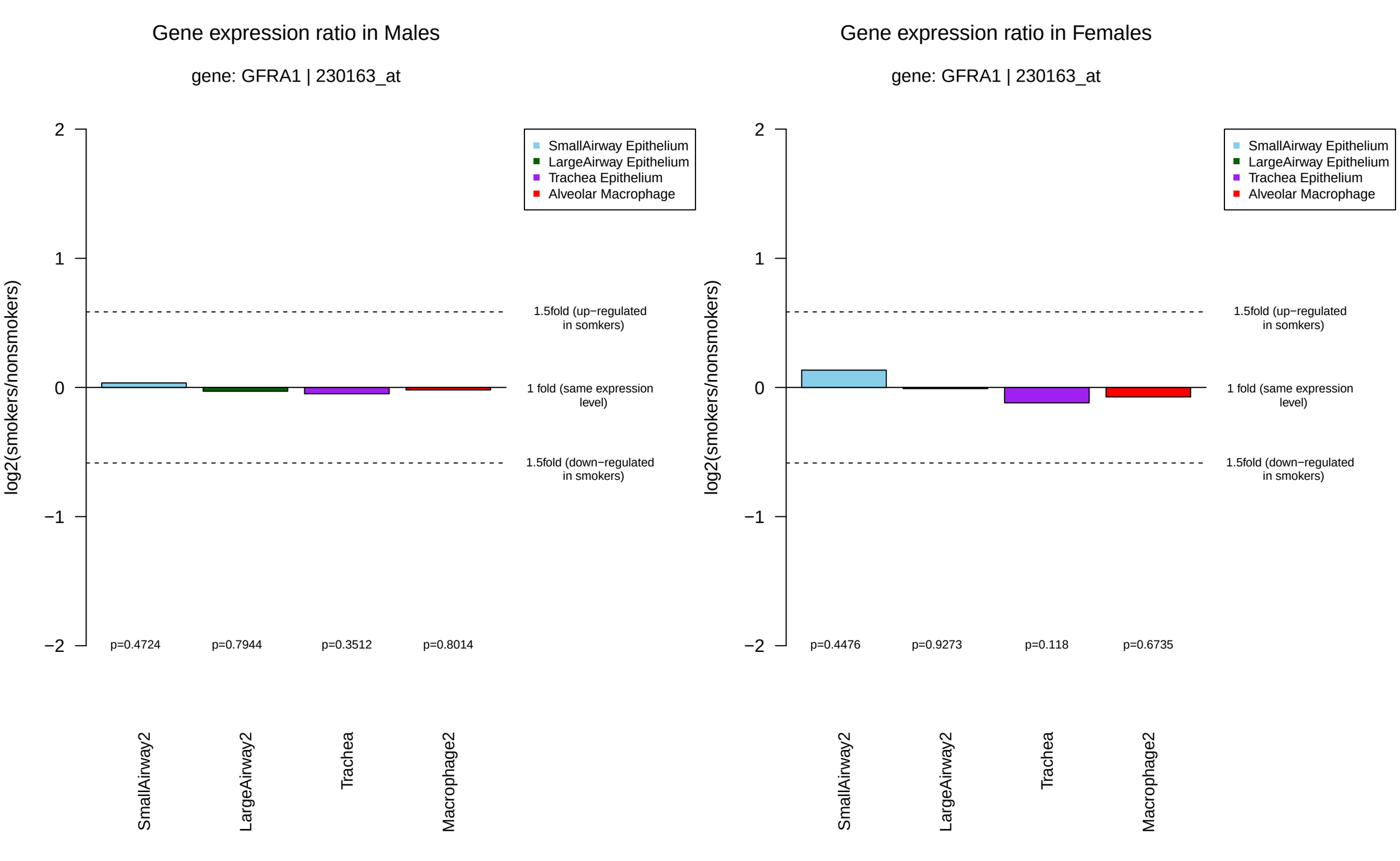
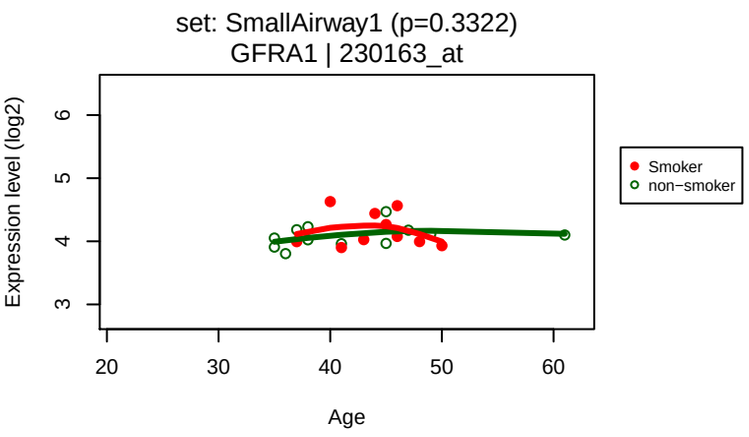
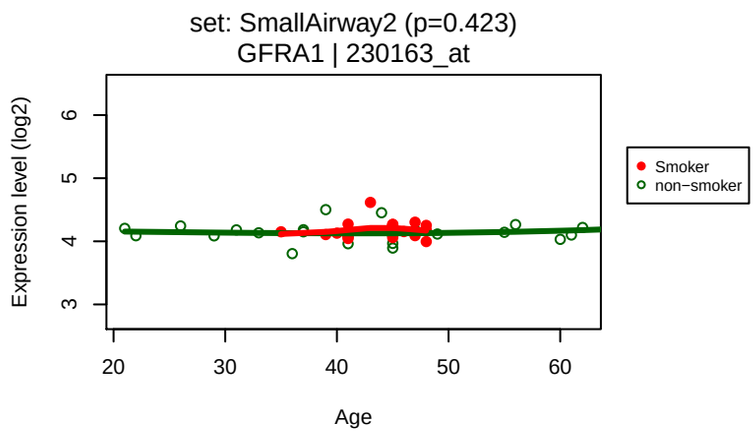
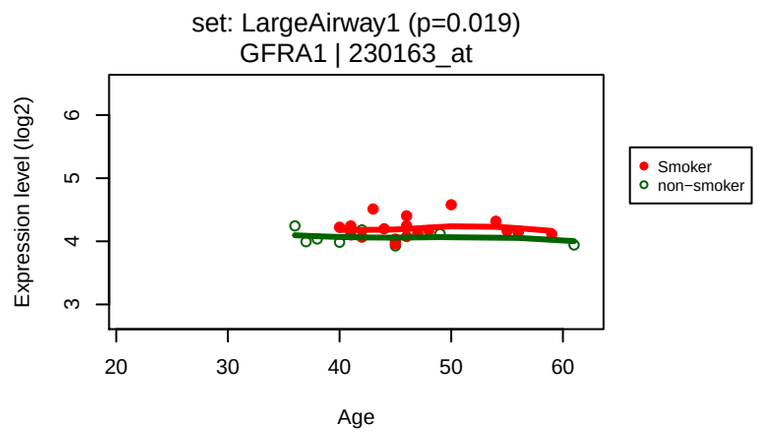

Gene expression of smokers and nonsmokers
gene: GFRA1 | 230163_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3322
p=0.423
p=0.1631
p=0.019
p=0.8768
p=0.1007
p=0.553
p=0.5829
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

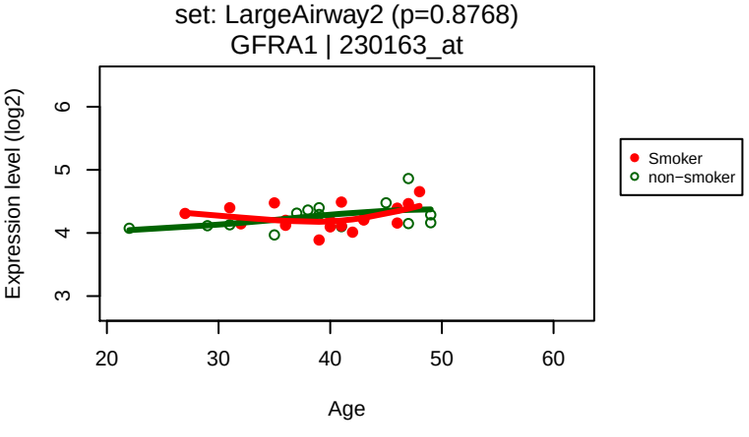

Gene expression ratio (smokers/nonsmokers)
gene: GFRA1 | 230163_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3322
p=0.423
p=0.1631
p=0.019
p=0.8768
p=0.1007
p=0.553
p=0.5829
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

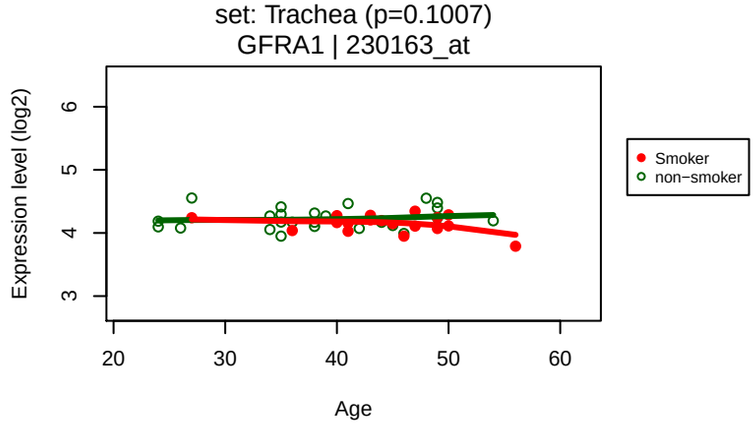

Gene expression ratio in Males
gene: GFRA1 | 230163_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4724
p=0.7944
p=0.3512
p=0.8014
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GFRA1 | 230163_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4476
p=0.9273
p=0.118
p=0.6735
SmallAirway2
LargeAirway2
Trachea
Macrophage2

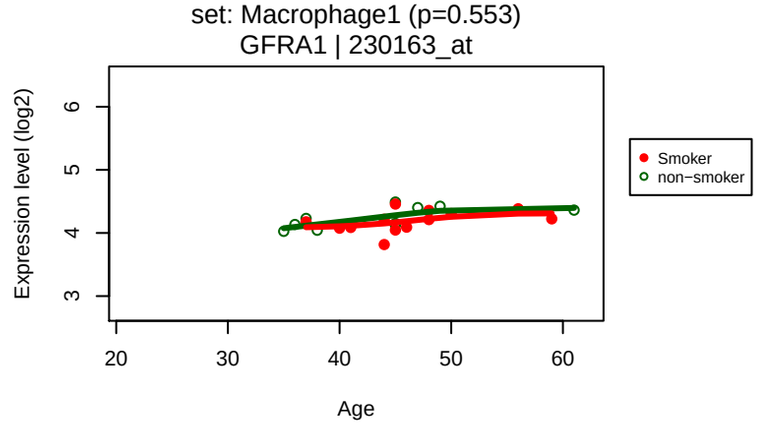

set: SmallAirway1 (p=0.3322)
GFRA1 | 230163_at
Expression level (log2)
Age
Smoker
non−smoker

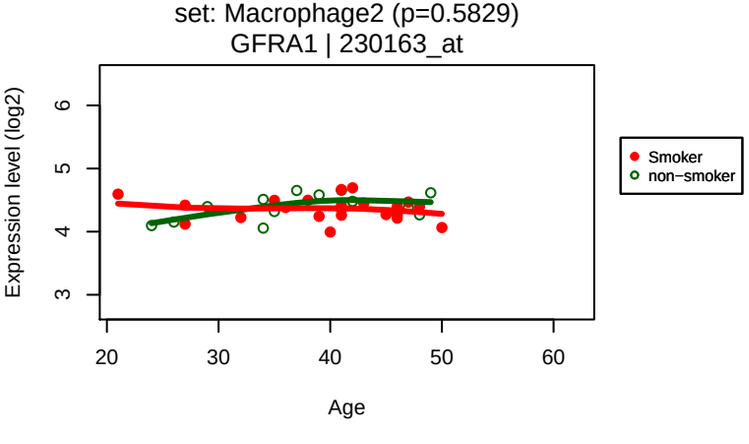

set: SmallAirway2 (p=0.423)
GFRA1 | 230163_at
Expression level (log2)
Age
Smoker
non−smoker

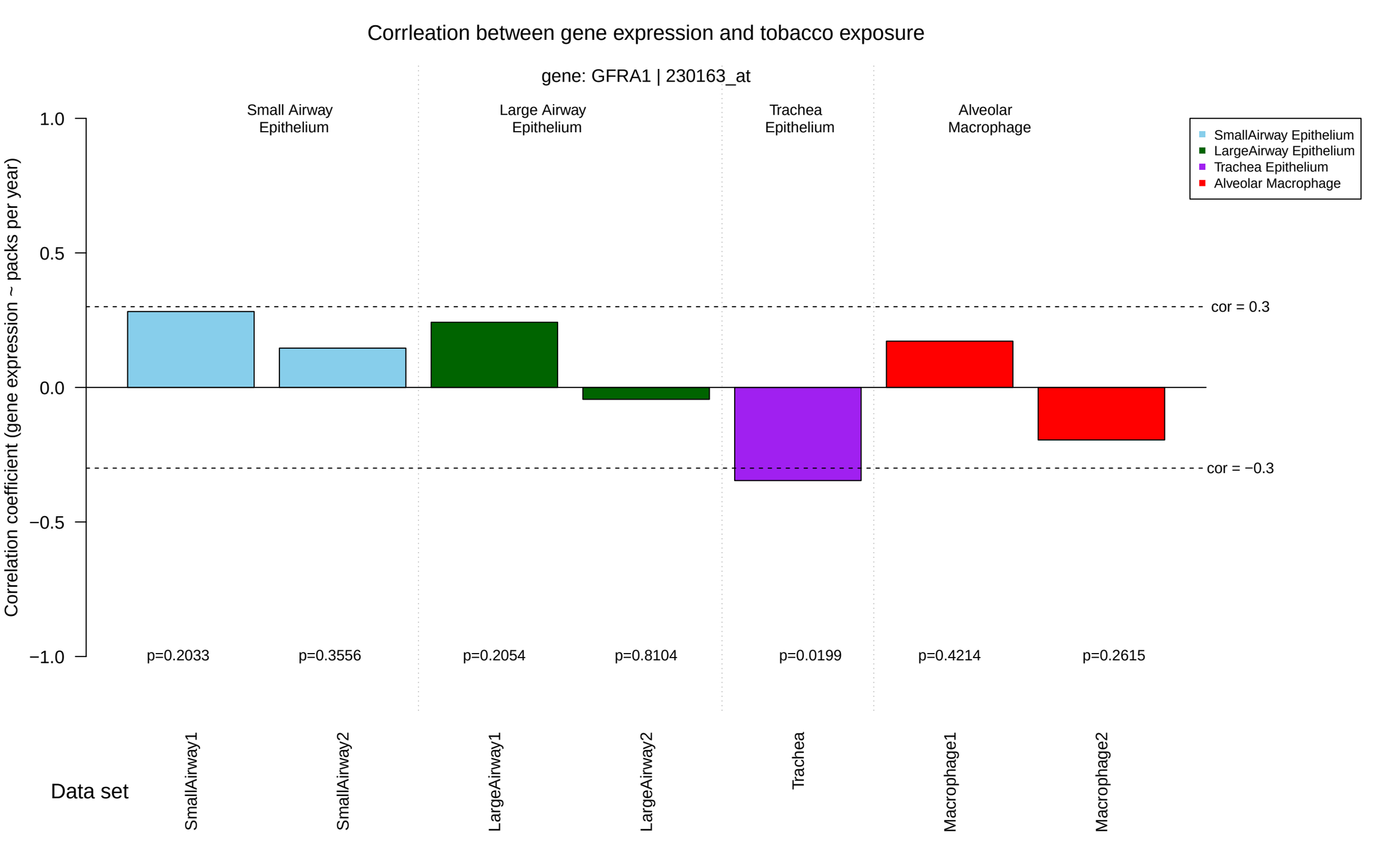

set: LargeAirway1 (p=0.019)
GFRA1 | 230163_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8768)
GFRA1 | 230163_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1007)
GFRA1 | 230163_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.553)
GFRA1 | 230163_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5829)
GFRA1 | 230163_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GFRA1 | 230163_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2033
p=0.3556
p=0.2054
p=0.8104
p=0.0199
p=0.4214
p=0.2615
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2